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SyncDreamer: Controllable and Expressive Avatar Generation Beyond the Talking Head

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Abstract

Generating realistic and expressive audio-driven talking avatars remains a challenge in digital human synthesis. Existing methods often depend on intermediate representations for natural body motion, restricting flexibility and leading to visual distortions. Moreover, many approaches rely on discrete emotion labels to regulate expression. Such categorical supervision fails to capture the continuous and fine-grained speech dynamics (rhythm, energy, intensity), resulting in limited synchronization and emotionally shallow motion. To overcome these limitations, we present ***SyncDreamer****, a unified diffusion Transformer framework that generates identity-preserving and emotionally expressive talking avatars from only a single image, speech audio, and text prompt. We propose a visual adapter with* ***Attention Localization Loss*** *to maintain identity fidelity, further incorporating an audio dynamics encoder for rhythm- and emotion-aware motion, and an RL-based* ***Cross-Modal Prompt Enhancer*** *grounding textual cues in visual context for fine-grained motion control. Extensive experiments on portrait and full-body benchmarks demonstrate state-of-the-art performance in realism, synchronization accuracy, and semantic controllability, establishing a scalable foundation for expressive digital avatars in interactive and creative applications.* https://fnazarieh.github.io/SyncDreamerWeb/

1. Introduction

Generating realistic and expressive talking avatars is a core objective in digital human research. Among its core sub-

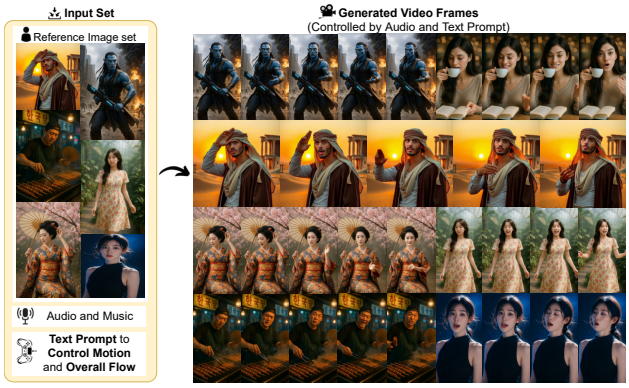


Figure 1. **Overview of SyncDreamer results.** From a reference image, audio, and text prompt, **SyncDreamer** generates realistic and identity-consistent talking avatars. **SyncDreamer** enables controllable and expressive motion generation, achieving synchronized lip movements, gestures, and gaze dynamics that align with both the audio and textual intent.

fields, talking portrait generation aims at synthesizing high-fidelity facial motion from audio and other input signals. This capability underlines a wide range of applications such as virtual assistants, content creation, film, and virtual presenters [30]. With the rapid evolution of multimodal generative models and interactive environments, there is an increasing demand for avatars that move and speak in harmony with linguistic content and emotions, enabling more natural and engaging human-computer interaction.

For clarity, we consider two representative settings of audio-driven avatar generation: (1) *portrait-based methods*, which synthesize facial expressions, lip motion, and modest head movement within a head–shoulders crop, and (2) *full-body methods* that extend motion generation to upper or full-body kinematics for synchronized gesture and posture control. Portrait-based models [8, 36, 50, 58] achieve

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acceptable audio–lip synchronization and identity preservation, yet focus primarily on mouth–speech correlation and lack explicit modeling of natural head or gaze dynamics. Because head movement cannot be reliably inferred from audio alone, these methods typically rely on ground-truth poses from reference videos [16, 23, 41] or maintain a near-static head configuration [33, 52]. As a result, they often produce repetitive motion and limited expressiveness, with fixed head poses that reduce realism. Full-body approaches such as EchoMimicV2 [29] and CyberHost [27] support upper-body animation via external pose encoders, yet their dependence on 3D landmarks and motion trajectories limits motion flexibility and fine-grained control over gestures.

Beyond these structural constraints, existing audio-driven talking avatar models often rely on generic speech encoders [2, 37] that primarily capture linguistic content while overlooking expressive acoustic cues such as rhythm, vocal intensity, and non-verbal sounds. To mitigate, some works [28, 56] incorporate explicit emotion classifiers, but these are typically constrained to coarse, discrete categories (*e.g.*, happy, disgust, surprised), oversimplifying the rich and continuous nature of human speech. Such categorical supervision fails to model the continuous temporal variations and dynamic emotional transitions inherent to natural speech, music, or performance.

As a result, existing models often produce flat or temporally misaligned expressions that fail to match the emotional rhythm of speech. Although diffusion-based models exhibit strong temporal modeling, their controllability remains limited. In particular, current Diffusion Transformer (DiT) approaches [3, 9, 15, 35, 45] treat text merely as a descriptive prior, conveying background or style rather than body motion control signal. Therefore, adding explicit cues (*e.g.*, "turn left" or "raise the right hand") has little to no impact on generation. Consequently, fine-grained control over gestures, gaze, and head dynamics remains elusive, especially in complex or performance-driven scenarios. These challenges highlight the need for a unified multimodal framework that couples emotional audio dynamics, spatial identity coherence, and text-based behavioral control.

To address these limitations, we introduce **`SyncDreamer`**, a diffusion-based framework for generating expressive and identity-preserving talking avatars from audio, a single reference image, and natural language prompts. Built upon a Diffusion Transformer (DiT) backbone, **`SyncDreamer`** integrates novel specialized modules for visual identity preservation, audio-driven motion dynamics, and text-guided behavioral control, enabling coherent, emotionally grounded, and user-controllable generation across both portrait and full-body scenarios.

To maintain identity consistency and preserve visual appearance in a video sequence, we introduce a *Visual Adapter* with an *Attention Localization Loss* (§ 3.1). The adapter distills identity-aware embeddings from the reference image via a learnable query mechanism that selectively attends to visual tokens, enabling the model to retain fine-grained appearance details during generation. Meanwhile, the loss guides cross-attention toward key semantic regions (*e.g.*, eyes, mouth, hands), enhancing spatial consistency and mitigating identity drift under large pose variations.

To capture expressive audio dynamics, we propose an *Audio Dynamics Encoder* that models fine-grained temporal variations in rhythm and energy by reweighting features based on expressive salience (§ 3.2). This enables **`SyncDreamer`** to produce temporally synchronized and emotionally coherent facial motions across diverse scenarios, from subtle conversation to performance-driven singing.

Finally, to overcome the limitations of pose or trajectory-based motion control, we introduce a *Cross-Modal Prompt Enhancer* that converts natural language prompts into active body motion control signals (§ 3.3). The module extracts visual cues *(e.g.*, attire, posture, background) from the reference image and refines user text via a vision–language model with chain-of-thought reasoning. Fine-tuned using Group Relative Policy Optimization (GRPO) [40] and novel task-specific rewards for visual grounding, gesture clarity, and temporal coherence, it transforms underspecified prompts into motion-aware instructions, enabling precise, context-aligned behaviors such as "delivering a talk with confident gestures" or "singing with rhythmic arm movements."

Together, these modules establish a unified multimodal diffusion framework that integrates appearance and identity preservation, expressive and emotionally coherent motion modeling, and fine-grained, text-driven behavioral control. Our main contributions are as follows:

- We propose **`SyncDreamer`**, a Diffusion Transformer-based framework for expressive avatar generation from a single reference image, audio, and natural language prompt, for both full-body and facial close-ups.
- We introduce three key modules: a *Visual Adapter* with an *Attention Localization Loss* for preserving visual identity, an *Audio Dynamics Encoder* for capturing expressive temporal cues from audio, and an *Cross-Modal Prompt Enhancer* that enables fine-grained, text-driven motion control grounded in visual context.
- Extensive evaluations on portrait and full-body benchmarks show that our method achieves state-of-the-art performance in realism, synchronization, expression alignment, identity preservation, motion controllability and smoothness.

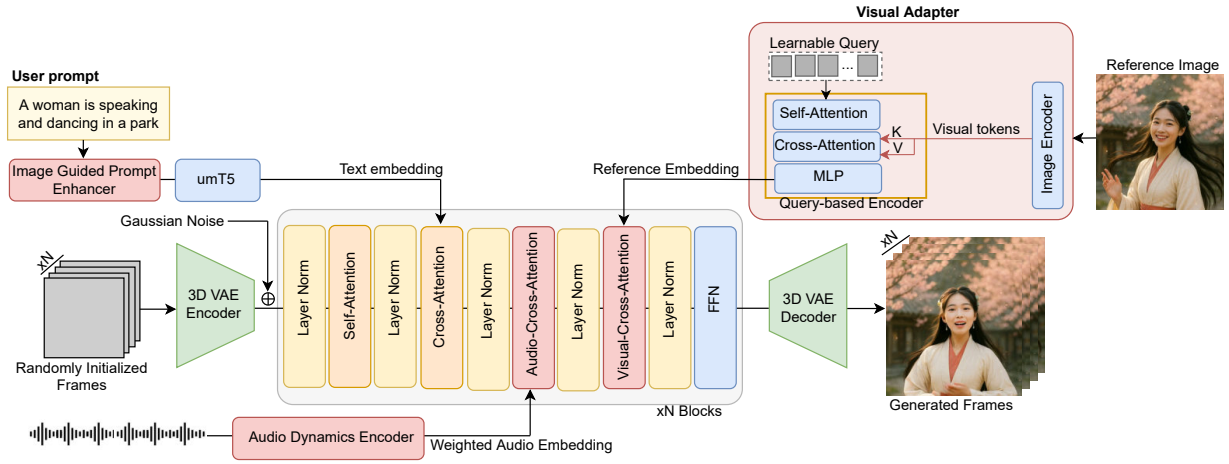


Figure 2. **Overview of the `SyncDreamer` architecture.** Given a reference image, audio, and a text prompt, **`SyncDreamer`** generates expressive, identity-preserving talking avatar videos with synchronized audio-lip motion and full-body dynamics.

2. Related Work

2.1. Video Diffusion Models

Recent progress in video generation has built upon the foundations of diffusion models [11], advancing temporal consistency and visual fidelity. Early works such as AnimateDiff [17] and Stable Video Diffusion [19] extended 2D diffusion models by introducing temporal layers to ensure frame-to-frame coherence. Later, Transformer-based designs unified spatial and temporal dependencies through joint attention mechanisms. Recent DiT variants like VACE [24] and TORA [55] incorporate multi-modal and trajectory-aware conditioning for controllable video generation. Large-scale systems such as Wan [43], Sora [57], and HunyuanVideo Avatar [3] also demonstrate DiT scalability for high-fidelity video synthesis. Building on recent advances, **`SyncDreamer`** introduces new conditioning modules to enable fine-grained motion control in avatar generation.

2.2. Audio-Driven Talking Face Generation

Early audio-driven talking face models used GANs [36, 58] or 3DMMs [14, 18, 39] to achieve audio–lip synchronization, but often lacked expressiveness and temporal coherence. Diffusion-based methods [4, 8, 12] improved realism yet remained head-centric, producing limited head and gesture motion. Recent extensions explore semi- or full-body generation. EchoMimicV2 [29] introduces detailed hand and finger landmarks for upper-body animation but depends heavily on predefined motion inputs. Hallo3 [9] adds text as auxiliary conditioning, though it primarily aids scene reconstruction rather than motion control. More advanced DiT-based models [3, 15, 45] leverage large-scale pretraining and multimodal conditioning for high-fidelity, synchronized avatars, but still treat text as a static style or context cue rather than an actionable motion directive. In contrast, our framework employs language as an active motion-control signal, aligning user intent, speech rhythm, and visual identity to generate semantically grounded, expressive full-body behavior.

3. Methodology

Given a single reference image, speech audio, and a text prompt, **`SyncDreamer`** generates expressive and identity-preserving avatars with accurate audio–lip synchronization. In contrast to prior approaches, our model synthesizes dynamic and realistic full-body motion without relying on intermediate representations such as 3D landmarks or pose sequences. An overview of the proposed framework is shown in Figure 2. The following sections describe the key components of our architecture and their roles in enabling visually grounded, expressive, and controllable avatar generation.

3.1. Visual Adapter

Ensuring a consistent visual appearance in audio-driven avatars is challenging, especially under expressive motion or large head rotations, where appearance drift often occurs. To address this, we propose a Visual Adapter that extracts a compact, identity-aware representation from a single reference image.

As illustrated in Figure 2, the Visual Adapter comprises two main components: an Image Encoder, and a Query-Based Encoder. The reference image is first processed by image encoder to obtain a sequence of visual tokens encoding high-level semantic and appearance features. To derive a concise and identity-focused representation, these visual tokens are then processed by a Query-Based Encoder akin to the Q-Former architecture [1, 26]. A set of *learnable query embeddings* attends to the visual tokens through cross-attention, allowing the model to selectively aggregate

information most relevant to subject's identity (*e.g.*, facial features, clothing) and background visual context. This process yields a set of *reference embeddings* capturing essential visual cues needed for appearance preservation.

To embed appearance information into the generative process, we introduce a *Visual Cross-Attention Block* within the Transformers backbone. It allows the generative model to attend to the reference embeddings during generation, guiding the output towards target identity. It is challenging for existing systems to maintain spatial alignment in long or expressive sequences, causing attention drift and misplacement of fine facial details. We address this by proposing an *Attention Localization Loss*(§ 3.1.1) that regularizes cross-attention to focus on key semantic regions (*e.g.*, eyes, hair, jawline), improving spatial consistency and visual fidelity.

3.1.1. Attention Localization Loss

Maintaining spatial alignment between the generated output and the reference image becomes challenging given dynamic motion or long video sequences. To address this, we introduce an Attention Localization Loss that regularizes the diffusion model's cross-attention behavior. The loss encourages each spatial query within the diffusion model to focus on reference embeddings corresponding to semantically related regions in the reference image. For instance, latent tokens near the mouth should primarily attend to mouth-related features, ensuring region-specific correspondence during generation.

To provide supervision, we obtain semantic region masks for key facial components in the reference image using segmentation models [38, 51]. Although the reference embeddings produced by the Visual Adapter are not spatially anchored, we apply soft supervision by penalizing attention weights that extend beyond their relevant regions. This encourages compact, semantically coherent attention patterns and improves identity consistency across frames. Formally, let $\mathbf{A}_{ij}$ denote the normalized attention weight between the i-th spatial latent token (query) and the j-th reference embedding token (key). We define a binary supervision mask $M_{ij} \in \{0, 1\}$, where $M_{ij} = 1$ if the spatial position of query i corresponds to the semantic region represented by reference token j, and 0 otherwise. The Attention Localization Loss is formulated as:

$$\mathcal{L}_{\text{loc}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{K} \mathbf{A}_{ij}(1 - M_{ij}), \tag{1}$$

where N and K denote the number of spatial latent and reference tokens, respectively. Minimizing $\mathcal{L}_{\text{loc}}$ encourages the model to penalize diffuse attention distributions over unrelated regions, promoting more localized and semantically meaningful correspondence. The final training objective combines the regularization with the standard diffusion reconstruction loss:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_{\text{loc}} \cdot \mathcal{L}_{\text{loc}}, \tag{2}$$

where $\lambda_{\text{loc}} = 0.4$ and controls the trade-off between generative quality and spatial precision. We apply this supervision to mid-level attention layers, where identity and semantic features are most discriminative. Aligning cross-attention distributions with key facial regions substantially improves spatial coherence and identity preservation, particularly for large head rotations, occlusions, and extended motion sequences.

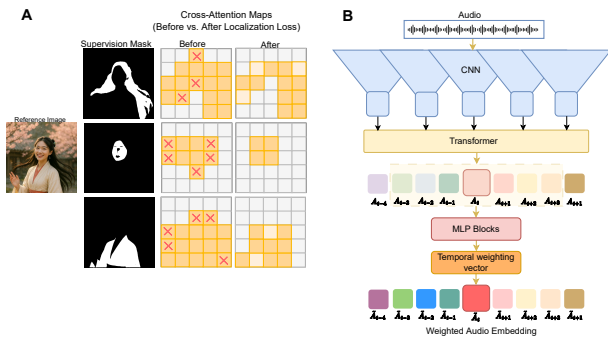


Figure 3. **(A) Attention Localization Loss.** Regularizes visual cross-attention maps within the **SyncDreamer** backbone to maintain spatial alignment between generated content and reference features. **(B) Audio Dynamics Encoder.** Refines audio features using a temporal weighting mechanism that emphasizes motion-relevant acoustic cues such as rhythm and vocal intensity.

3.2. Audio Dynamics Encoder

Generating expressive and synchronized facial motion requires capturing fine-grained acoustic cues such as rhythm, energy, and emotion. However, conventional speech encoders (*e.g.*, Wav2Vec2, Whisper [2, 37]) focus on linguistic content, overlooking these expressive dynamics. Prior attempts [28, 56] to model expressiveness often rely on discrete emotion or motion labels, failing to model the continuous spectrum of human expressivity, leading to flat or unsynchronized motion.

To address these limitations, we propose an Audio Dynamics Encoder that captures motion-relevant acoustic cues from speech and singing. Specifically, it extends Wav2Vec2 with a temporal context module that employs a sliding-window strategy to compute frame-level expressiveness scores. The encoder explicitly models temporal context and dynamically emphasizes expressive regions in the audio signal, enabling more synchronized, expressive, and emotionally coherent motion generation.

Let $A \in \mathbb{R}^{T \times d}$ denote the Wav2Vec2 feature sequence, where T is the number of time steps and d the feature dimension. For each audio frame t, a local temporal window $A_{t-k:t+k} \in \mathbb{R}^{(2k+1) \times d}$ is processed by a temporal MLP module f_{temp}, which aggregates contextual information and

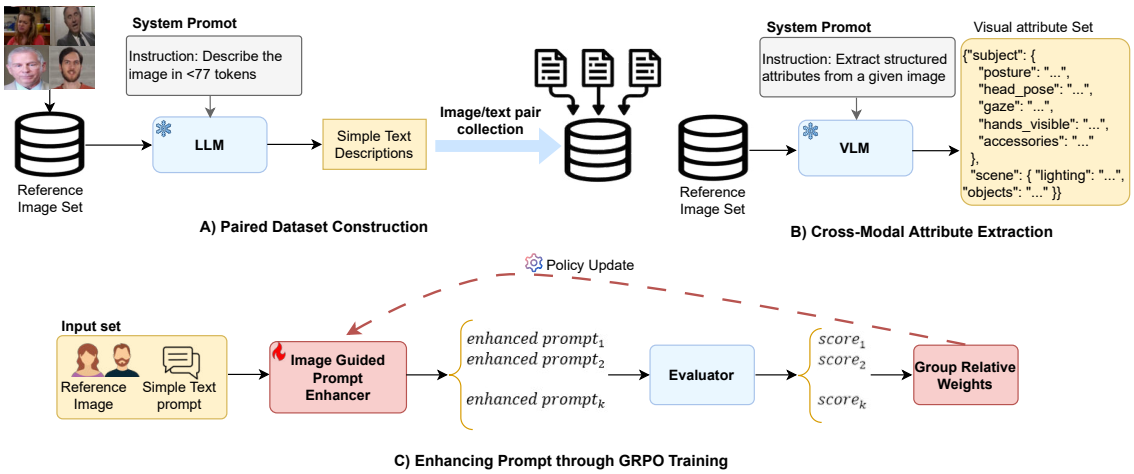


Figure 4. **Cross-Modal Prompt Enhancer**. **SyncDreamer** augments textual prompts with visual cues extracted from a reference image. *(A) Pair Dataset Construction*: Image–text pairs are collected using LLM-based captioning. *(B) Cross-Modal Attribute Extraction*: A VLM extracts structured semantic attributes from reference images. *(C) Prompt Enhancement*: enriched prompts are generated and optimized through Group Relative Policy Optimization, guided by a reward that evaluates visual relevance, motion specificity, and fluency.

produces a scalar weight:

$$w_t = f_{\text{temp}}(A_{t-k:t+k}), \quad w_t \in [0, 1]. \tag{3}$$

Using a fixed window size of $k = 3$ (three frames before and after each timestep), we obtain a temporal weighting vector $w = [w_1, \ldots, w_T]$, where each w_t reflects the expressive salience of the corresponding audio frame. The reweighted features are then computed as:

$$\tilde{A}_t = w_t \cdot A_t, \quad \forall t \in \{1, \ldots, T\}. \tag{4}$$

This formulation amplifies regions of high expressive content, such as rhythmic accents, intensity peaks, or emotional transitions, while suppressing flat or silent segments. Empirically, replacing the proposed Audio Dynamics Encoder with unweighted Wav2Vec2 features leads to reduced motion expressiveness and weaker audio–visual synchronization (see § 5.3 for ablation results).

3.3. Cross-Modal Prompt Enhancer

Existing audio-driven talking avatar models rely on a reference image and speech to generate audio–lip synchronized motion, occasionally incorporating textual descriptions as auxiliary conditioning. In most systems [9, 31, 32, 45], text serves only as a passive descriptor, providing contextual or stylistic cues rather than functioning as an active motion control signal. Consequently, users have limited influence over dynamic behaviors such as gestures, head movements, and gaze shifts. Moreover, user-provided prompts are often brief or underspecified, leading to weakly aligned motion.

To address this limitation, we first construct a paired dataset including reference images and their corresponding textual descriptions. Since standard talking face datasets contain only video and audio streams, we extract a single frame from each video as the reference image and employ an LLM [7] to generate a brief (fewer than 77 tokens) and semantically aligned caption describing its visual content. These captions capture coarse visual attributes, such as clothing and background context, providing the linguistic grounding necessary for training the proposed prompt enhancement module. **SyncDreamer** utilizes this dataset for training a vision-language Model (VLM) to augment user-provided prompts with visual cues derived from the reference image (Fig. 4). Specifically, this work uses the Qwen2-VL model [46] to extract semantic attributes including attire, background, objects, body posture, and implied motion (*e.g.*, "tilting the head" or "holding a pen"). These attributes are integrated with the original user prompt to produce a meaningful and motion-aware instruction to guide the avatar generation process.

The model is optimized using Group Relative Policy Optimization (GRPO), a ranking-based reinforcement learning algorithm. For each input, the enhancer generates multiple candidate prompts that are ranked by a task specific reward function assessing visual relevance, motion specificity, and language fluency. GRPO updates the model to prefer candidates yielding more grounded and coherent generations, without the need for an explicitly learned reward model. The enhanced prompt is encoded using a T5 text encoder [5] and used as conditioning for body motion generation, enabling visually grounded control over gestures, body turns, and gaze dynamics. To ensure the enhanced prompts contribute effectively to motion control, we further design a task-specific scoring function that evaluates their quality across several dimensions relevant to expressive and semantically aligned avatar motion. *Detailed explanation*

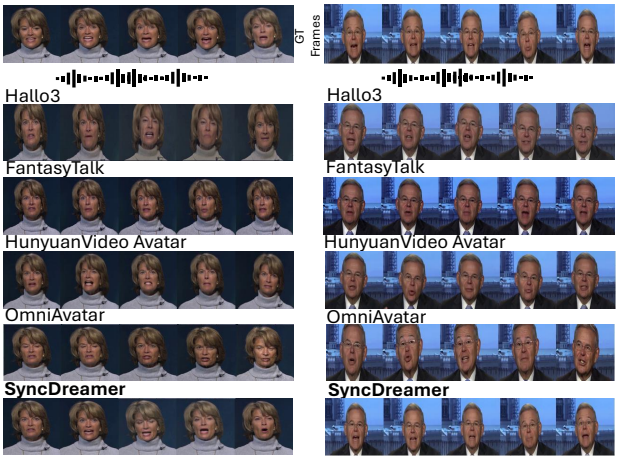


Figure 5. **Qualitative comparison with prior methods.** **`SyncDreamer`** generates facial outputs with consistent identity, smooth motion transitions, and precise audio–lip synchronization.

Table 1. Quantitative comparison on the HDTF dataset. Our model achieves the best performance across all key metrics, including perceptual quality, lip-sync accuracy, and identity fidelity.

Method	FID ↓	FVD ↓	Sync-C ↑	Sync-D ↓	IQA ↑
V-Express [44]	62.8	972.3	4.13	10.55	3.14
Hallo [50]	62.1	715.8	6.81	7.83	3.26
Hallo2 [8]	61.5	631.9	6.87	7.16	3.35
EchoMimic [4]	59.8	581.6	7.15	6.91	3.41
Hallo3 [9]	56.4	579.1	7.19	6.73	3.47
FantasyTalk [45]	62.7	594.6	5.48	9.21	3.21
HunyuanVideo Avatar [3]	54.9	528.3	7.34	6.48	3.56
OmniAvatar [15]	53.7	514.7	7.92	6.36	3.68
`SyncDreamer`	**52.8**	**508.1**	**8.04**	**6.15**	**3.72**

on reward formulations is provided in supplementary materials.

4. Implementation Details

Our framework is built upon the diffusion transformer [43] backbone and is trained in two stages. In **Stage 1**, the model is fine-tuned on text–video pairs from HumanVID to learn motion and scene composition priors. In **Stage 2**, audio conditioning is introduced using Hallo3, HDTF, and AVSpeech for synchronized, expressive motion generation. To improve robustness, we apply classifier-free guidance by randomly dropping conditioning modalities (audio, image, or text) with a 0.1 probability during training. Our model is trained using two A100 GPUs and consists of 42 Transformer blocks. Training and inference use 100 denoising steps to synthesize identity-consistent, rhythm-aligned avatars. For evaluation, we test on both real and synthetic reference images generated with GPT-4o [34] and Flux [25], ensuring no training overlap. *Further training configurations, and additional results are included in the supplementary material.*

4.1. Datasets and Evaluation Metrics

We construct a large-scale dataset by aggregating diverse, publicly available audio–visual sources capturing varied



Guiding textual Description:

Modern office with floor-to-ceiling windows showing city buildings, open wooden shelves with books and plant pots, a light-wood desk. The subject wears a navy suit over a light blue shirt. He holds a silver tablet in his left hand. the right hand is free.
Beginning: Standing, facing the camera. **left hand holding the tablet.**
Middle: **The right hand makes small, smooth gestures (open-palm emphasis, short arcs) while the left hand keeps the tablet steady.** Occasional slight head nods. gaze mostly at the camera. Torso stable.
End: The right hand returns to a relaxed position. **The tablet stays in the left hand.** He finishes looking into the camera with a **professional, friendly expression.**

Figure 6. **Qualitative comparison on upper-body talking avatars.** Baselines fail to follow motion cues, yielding weak gesture control, whereas **`SyncDreamer`** generates semantically aligned, coherent upper-body motion with consistent object handling and rhythm-aware gestures.

motion styles, identities, and viewpoints. The data is categorized into three framing types, *full-body*, *upper-body*, and *facial close-ups*, enabling expressive avatar synthesis across speaking and performance scenarios. Specifically, full-body data is drawn from **HumanVID** [48] (19K YouTube videos), upper-body clips from **Hallo3** [9] (10K expressive samples), and facial close-ups from **HDTF** [54] and **AVSpeech** [13]. We ensure quality through SyncNet-based [6] audio–visual alignment and InsightFace [10] filtering, retaining 650K single-speaker clips (5–30s) standardized in resolution. This dataset supports training ex-



Figure 7. **Motion editing.** Given a reference image, audio and text prompt for body motion control, **SyncDreamer** generates motion aligned with text prompts, producing expressive dance, adaptive gestures, and semantically consistent actions, while maintaining identity and speech synchronization.



Figure 8. **Qualitative comparison showing the impact of Attention Localization.** Our method maintains identity and spatial consistency, while the baseline shows facial drift and feature misplacement.

pressive, identity-consistent avatars across diverse emotional, musical, and performance-driven contexts.

We evaluate performance using perceptual, temporal, and synchronization metrics. Fréchet Inception Distance (FID) [20] and Fréchet Video Distance (FVD) [42] assess visual realism and temporal coherence. SyncNet Confidence (Sync-C) and Distance (Sync-D) [6] measure audio–lip alignment. PSNR [21], SSIM [47], CSIM, and IQA [49] evaluate frame fidelity and identity preservation.

5. Results

5.1. Quantitative and Qualitative Results

We compare **SyncDreamer** with recent audio-driven avatar models, including classical approaches [4, 44, 50] and diffusion Transformer-based systems [3, 9, 15, 45]. Quantitative results on HDTF and AVSpeech (Table 1, 2) show that **SyncDreamer** consistently outperforms all baselines in visual realism, synchronization accuracy, and identity preservation. These improvements stem from the complementary design of our key modules: the Visual Adapter and Attention Localization Loss jointly maintain spatial and identity consistency under expressive motion, while the Audio Dynamics Encoder captures fine-grained temporal variations in rhythm and vocal intensity, resulting in more natural, emotionally grounded, and temporally coherent motion. Qualitative comparisons (Figure 5) further confirm that **SyncDreamer** produces smoother motion, sharper details, and more stable gaze and lip alignment than prior methods.

Table 2. Quantitative results on the AVSpeech benchmark. Our method shows better performance across most metrics, demonstrating strong generalization and robustness in in-the-wild, speech-driven facial synthesis.

Method	FID ↓	FVD ↓	Sync-C ↑	Sync-D ↓	IQA ↑
V-Express [44]	89.5	1531	3.74	9.15	2.48
Hallo [50]	79.1	104.6	5.64	8.91	2.51
Hallo2 [8]	74.9	87.5	5.12	8.31	2.57
EchoMimic [4]	71.5	83.1	5.33	8.14	2.61
Hallo3 [9]	71.8	81.9	5.86	7.28	2.66
FantasyTalk [45]	83.6	91.3	4.07	8.58	2.43
HunyuanVideo Avatar [3]	69.4	79.5	5.91	7.21	2.71
OmniAvatar [15]	68.4	73.8	6.24	**6.73**	2.78
SyncDreamer	**67.9**	**72.6**	**6.31**	7.05	**2.81**

Table 3. Quantitative comparison on the EMTD benchmark. Our model achieves the best performance across all metrics, demonstrating superior visual fidelity, temporal consistency, and structural alignment in full-body talking avatar generation.

Method	FID ↓	FVD ↓	PSNR ↑	SSIM ↑	CSIM ↑
AnimateAnyone [22]	56.31	871.36	20.63	0.713	0.269
MimicMotion [53]	51.82	614.98	19.71	0.725	0.371
EchoMimicV2 [29]	42.69	528.10	22.09	0.741	0.393
SyncDreamer	**41.77**	**483.59**	**22.38**	**0.746**	**0.598**

5.2. Generalization Beyond Portraits

Quantitative comparison. Most existing benchmarks for audio-driven talking faces focus on headshots, limiting evaluation to facial motion. To assess broader generalization, we evaluate on EMTD [29] benchmark, which includes upper-body speaking with diverse gestures. As shown in Table 3, **SyncDreamer** outperforms state-of-the-art pose-driven models [22, 29, 53] across all metrics. Unlike prior methods relying on pose representations, **SyncDreamer** learns motion end-to-end from audio and text, enabling more coherent and controllable full-body generation without handcrafted constraints.

Qualitative Comparison. Figure 6 compares **SyncDreamer** with recent Diffusion Transformer-based models on upper-body speaking scenarios. Prior works either rely on motion trajectories [22, 29, 53], limiting flexibility, or use text only for scene description [3, 9, 15, 45], not body motion control. When

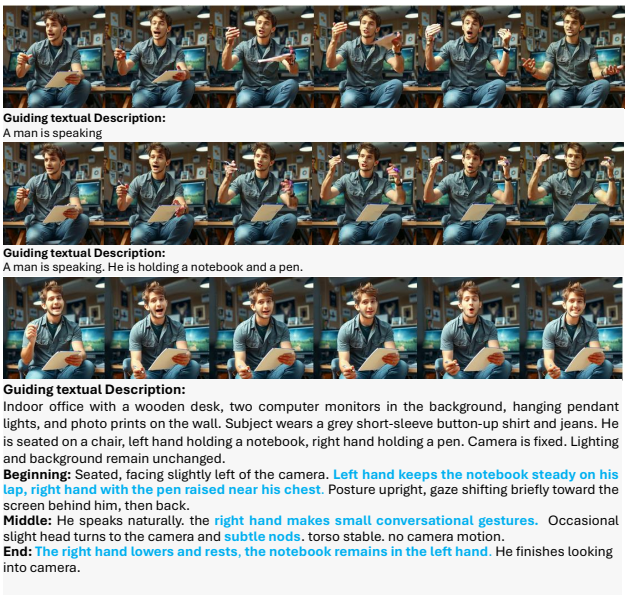


Figure 9. **Effect of the Prompt Enhancer.** Compared with minimal or manually refined prompts, our enhanced prompts yield more natural gestures, preserve object continuity, and maintain stronger identity–scene coherence across the sequence.

prompted with our enhanced instructions (§ 3.3), these baselines fail to align gestures and gaze with semantic intent. For example, in Figure 6 (row 4), the tablet is abruptly transferred between hands mid-sequence, revealing a lack of temporal consistency and visual grounding. As shown in Figure 7, **SyncDreamer** extends this capability to diverse scenarios, generating motion that coherently follows both audio rhythm and textual intent. It performs complex, context-aware behaviors, such as dancing (row 1), adjusting body orientation (row 2), or executing head and hand movements (row 3), directly from text, without pose templates or motion trajectories. Notably, in the third row, despite a downward-facing reference image, **SyncDreamer** reconstructs natural head-lifting and maintains the identity details throughout the video. This demonstrates **SyncDreamer** 's flexibility for semantically grounded motion control across expressive full-body and performance-driven settings.

5.3. Ablation Study

Effect of Attention Localization Loss We ablate the proposed *Attention Localization Loss* to evaluate its contribution to spatial consistency and identity preservation. As shown in Figure 8, removing this component results in noticeable identity drift, including misaligned facial features and displaced visual elements, particularly during expressive movements such as page turning or hand gestures. By encouraging cross-attention to focus on semantically relevant regions, the loss serves as a spatial prior that reinforces stable identity features and prevents degradation of fine-grained visual details over time.

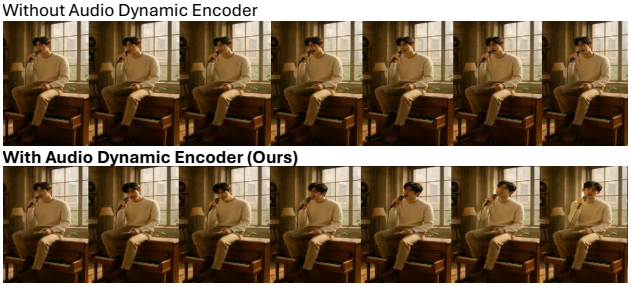


Figure 10. **Effect of the Audio Dynamics Encoder.** Although aligning facial and body motion with audio is challenging, particularly in full-body cases, our model naturally adapts motion to the dynamics of the speech signal.

Effect of the Audio Dynamics Encoder To assess the role of the *Audio Dynamics Encoder*, we perform ablations on expressive and non-verbal audio scenarios (*e.g.*, singing, humming). As illustrated in Figures 10, omitting this module leads to flattened expressions and diminished emotional fidelity, especially during variations in rhythm or energy. In contrast, the full model produces synchronized and emotionally expressive behavior that adapts smoothly to acoustic dynamics. Compared to recent diffusion-based methods [3, 9, 15], which lack explicit modeling of non-verbal cues, our approach more effectively captures fine-grained temporal variations in both portrait and full-body settings.

Effect of the Cross-Modal Prompt Enhancer. We evaluate the influence of the *Cross-Modal Prompt Enhancer* by comparing three input conditions: (1) a minimal user prompt, (2) a manually enriched prompt, and (3) our automatically enhanced prompt. As shown in Figure 9, minimal prompts yield generic, weakly grounded motion with missing contextual elements (*e.g.*, disappearing notebook or pen). Manually extended prompts improve object persistence but still lack expressive behavior and visual coherence. In contrast, our enhanced prompts, augmented with image-grounded cues, generate avatars exhibiting context-aware gestures, stable object continuity, and consistent visual sharpness across frames.

6. Conclusion

We introduced **SyncDreamer**, a unified framework for audio-driven talking avatar generation that integrates appearance conditioning, expressive audio encoding, and cross-modal prompt enhancer. The system enables fine-grained, language-driven control of gestures, gaze, and full-body motion without explicit pose supervision. Extensive experiments demonstrate identity preservation, motion realism, and synchronization across diverse scenarios. **SyncDreamer** offers a foundation for controllable and expressive digital avatars in virtual assistance, education, and creative media.

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